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Practical Course Wearable Robotics
School of Computation and Technology

Practical course Wearable Robotics: Upper Limb Exoskeletons

Practical Report

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1 Introduction

Throughout the recent decades there has been a constantly growing interest in implementing exoskeleton technology as a new type of approach in various fields of life. All of these interests accumulate to the common goal of wanting to offer a bridge between human potential and technological innovation and being able to enhance one's own physical capabilities. Among its wide range of possible applications, exoskeletons designed for grasping actions hold particular promise, pushing the boundaries in the way humans interact with objects and environments. Most interactions that humans perform with their environment fall under the category of manipulation tasks, with grasping and picking up tasks being one of the the most prevalent chunks of it.
In the realm of healthcare, exoskeletons represent a beacon of hope for individuals recovering from injuries, disabilities or degenerative conditions, that limit their physical capabilities. By providing assistance and support during grasping movements, exoskeletons aim to empower patients to regain independence and functionality. Whether they are designed for use-cases such as picking up objects, using tools or performing every day activities, exoskeletons offer patients an opportunity to reclaim their autonomy and quality of life. Moreover, the personalized nature of exoskeletons enables tailored rehabilitation protocols, optimizing patient outcomes and accelerating the path to recovery.
Within the scope of being used in healthcare, this document proposes a design for an upper-body, arm-exoskeleton. The exoskeleton model is designed for grasping and picking-up actions and aims to reenable the patient to make full use of their arm and act as a support for the patient when executing the discussed activities.
The chapters were written under equal contribution of all members. The Introduction and the Mathematical Modelwere written by Richard Willke. The Use Case was written by Richard Willke and Nikolas Kaarsberg. Nikolas Kaarsberg also wrote the chapter about the Mehanic Model, which was proofread by Richard Willke and Lingfeng Gu. The work is being rounded up by the chapter about the Process Model and Controller Design, which was written by Lingfeng Gu. The presentation and some of the videos from the documentation are uploaded on the group's LRZ sync and share folder [1]. These include the final presentation as well as the tests that worked and did not work.

[1] https://syncandshare.lrz.de/getlink/fiT2JhfUk4qttPCsPKnpct/

The final tests, which yielded positive results, are located within the presentation folder.

2 State of the Art

In the current exoskeleton research environment, numerous advances are being made daily, from novel soft robotics solutions with tendons like the UBH-IV hand [2] to more conventional, conservative movement assistance such as with active rigid robotics [1] as well as passive mechanisms for static support [3]. The exoskeleton presented in this work is heavily inspired by a preexisting design, made by the Kyushu university in Japan. In such a rigid robotics approach is presented with a 7 DoF design with the aim to assist daily activities of physically weak persons such as elderly, disabled, or injured persons. Their system uses a combination of EMGs, angle sensors, stereo camera input and an ultrasonic sensor as sensory input to determine the system state. The robot assists the motion of the user based on the user's muscle activation, measured by the aforementioned EMG sensors. Tremor suppression methods are also studied [4]. Their design serves as an inspiration for our design of a solution for the support of the pronation and supination of the lower arm. The most noticeable aspect about their approach is the large gear through which the human arm is fit through. This design makes use of the gear's applied torque on the lower arm to achieve the rotational movement of the pronation and supination. The aim of the work presented in this document is the design of a 2 DoF movement-assisting exoskeleton prototype for a human arm. The initial approach is a simplified design of that presented by the Kyushu university, using only 2 DoF. The goal is to be able to assist the user in performing grasping actions as well as the use in physiotherapy, for example the relearning of specific movement actions through repetitive execution.

3 Design Approach

The task of this laboratory is the design of a two degree of freedom exoskeleton arm-piece, used to assist the user in grasping and picking-up actions. The two DoF of the human arm, that are the most used during said actions are the elbow flexion and extension, as well as the wrist pronation and supination. This chapter describes the different aspects of the exoskeleton design process, such as the calculation of the kinematics and dynamics through a mathematic model, the design of the exoskeleton model in CAD, the design of the controllers and assistive control modes as well as the sensor placement. The following sections of this chapter will go over each of them in further detail.

The materials that were used in the design process were one ESP32 microcontroller, programmed using the Arduino IDE, a laptop through which the controller was implemented, two servomotors as well as a 3D printer to create the plastic materials of the exoskeleton. An additional ball bearing has been ordered additionally. Its use is explained in further detail in section 4.3 .

3.1 Mathematic Model

The human arm can be modeled as a set of rotary joints. A simplified version of the human arm, using only 3 DOF is depicted in figure 3. For the modeling of this task, the latter two joints of the presented kinematic chain are considered. The remaining joints are seen as static in this case and thus not included in the calculation of the system dynamics of this project. All the calculations presented in this section were made using the formula provided in tutorial 3 of this project's course.

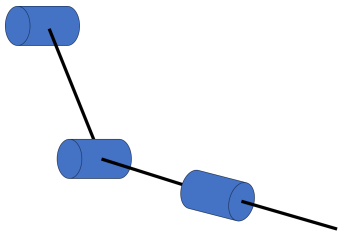


Figure 1: simplified model of the human arm using rotary joints

The relationship of the joints to each other is modeled using homogeneous transformations. The relationships of each joint to its predecessor is expressed in each of the matrices, as shown below. The above index always indicated the origin joint, and the lower index the goal joint that the transformation leads to. The last two matrices are the transformations of the centers of mass to each of their predecessor joint. The presented matrices can also be found in the analysejacobian.m file in the supplemented code.

$$T_1^0 = \begin{bmatrix} 1 & 0 & 0 & L_1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^1 = \begin{bmatrix} cos(q_1) & 0 & -sin(q_1) & 0 \\ sin(q_1) & 0 & cos(q_1) & 0 \\ 0 & -1 & 0 & -L_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^2 = \begin{bmatrix} cos(q_2) & -sin(q_2) & 0 & 0 \\ sin(q_2) & cos(q_2) & 0 & 0 \\ 0 & 0 & 1 & L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{cm,1}^1 = \begin{bmatrix} cos(q_1) & 0 & -sin(q_1) & 0 \\ sin(q_1) & 0 & cos(q_1) & 0 \\ 0 & -1 & 0 & -L_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{cm,2}^2 = \begin{bmatrix} cos(q_2) & -sin(q_2) & 0 & 0 \\ sin(q_2) & cos(q_2) & 0 & 0 \\ 0 & 0 & 1 & \frac{L_2}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The homogeneous transformations yield the configuration depicted in figure 2.

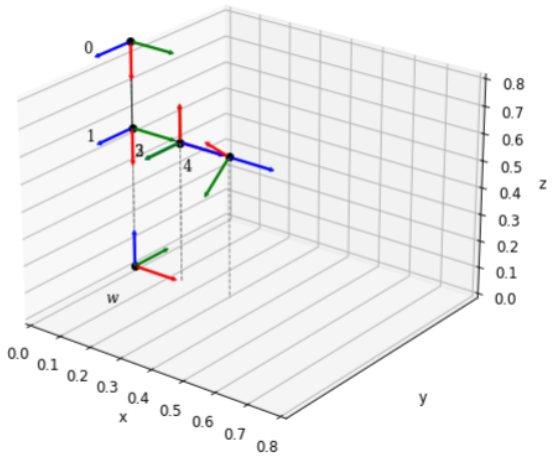


Figure 2: resulting model from the chain of homogenous transforms

The transforms are subsequently used to compute the Jacobian for the center of masses of the arm. Here, the only center of mass that is being moved is the mass m_2, which equals the center of mass of the lower arm, reaching from the base of the elbow to the base of the wrist.

$$J^1_{cm,2} = \begin{bmatrix} -\frac{L_2 cos(q_1)}{2} & \frac{L_2 cos(q_1)}{2} \\ -\frac{L_2 sin(q_1)}{2} & \frac{L_2 sin(q_1)}{2} \\ 0 & 0 \\ 0 & -sin(q_1) \\ 0 & cos(q_1) \\ 1 & 0 \end{bmatrix}$$

Using the Jacobian, the resulting matrices for the Inertia, Coriolis, Gravity and Friction are calculated, using the formulas as described in the tutorials of this course. For readability sake, sine and cosine are shortened to s and c respectively and their inputs as indices.

$$M = \begin{bmatrix} I_{133} + \frac{L_2^2 m_2}{4} + I_{222} c_{q_2}^2 + I_{211} s_{q_2}^2 + 2 I_{212} c_{q_2} s_{q_2} & I_{223} c_{q_2} - I_{213} s_{q_2} \\ -I_{223} c_{q_2} - I_{213} s_{q_2} & I_{233} \end{bmatrix}$$

$$C = \begin{bmatrix} q_{d,2}(\frac{2 I_{212} c_{2q_2} + (I_{211} - I_{222}) s_{2q_2}}{2}) & q_{d,1}(I_{212}(1 - 2 s_{q_2}^2) + (I_{211} - I_{222}) c_{q_2} s_{q_2}) - q_{d,2}(I_{213} c_{q_2} - I_{223} s_{q_2}) \\ -q_{d,1}(\frac{2 I_{212} c_{2q_2} + (I_{211} - I_{222}) s_{2q_2}}{2}) & 0 \end{bmatrix}$$

$$G = \begin{bmatrix} -\frac{L_2 m_2 (g_x c(q_1) + g_y s(q_1))}{2} \\ 0 \end{bmatrix}$$

$$B = \begin{bmatrix} b_1 \dot{q_1} & 0 \\ 0 & b_2 \dot{q_2} \end{bmatrix}$$

The non-q_x parameters are chosen either through measurements on the arm of one of the team members or through either rough estimation or choosing it arbitrarily. Since the design of the exoskeleton is still only a prototype, the parameters are chosen so that the exoskeleton can perform its tasks well. However, because of the little gain in performance that an optimization may pose, there is no optimization on these parameters done during the design of the exoskeleton.

3.2 Use Case

The system is designed for uses in rehabilitation of patients without neural impairments. The situations that are looked at in this scope are mainly for physiotherapy by either assisting the patient in their movement actions through continuous assistance control method or help them relearn specific movements and muscle control actions through repetitive movements using an activity triggered control method. To be able to change between modes easily, a switch was implemented on the breadboard where the ESP32 microcontroller is located.
Depending on the severity of the patients condition, or depending on the judgement of the physiotherapist, the physiotherapist can easily change between the two assistive methods with this switch. This allows them to more accurately decide which therapy method may be more suitable for the given patient in this use-case scenario.
The scenarios that are looked at are picking-up and holding tasks. With the exoskeleton, the patient shall be able to lift heavier objects more easily and also be able to hold them longer, in case they suffer from fast muscular fatigue.

3.3 Mechanic Model

This system takes heavy inspiration from a previous design constructed by the System Engineering Lab under the supervision of Prof. Kazou Kiguchi within the Faculty of Engineering of the Kyushu University. A photograph on the design, which is uploaded on the website of the institute, is depicted in figure 3.

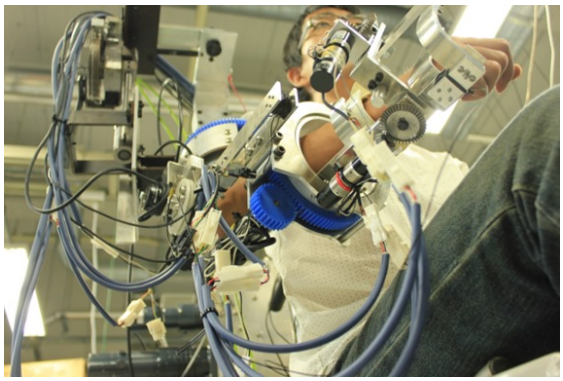


Figure 3: Kyushu prototype

The objective of the design presented in this document is to emulate the two main Degrees of Freedom (DoF) responsible for allowing grasping. Hence, a simplification from the Kyushu design to the two DoF that are the most impactful on grasping actions was possible.

The model's mechanics underwent several design phases, them being the design of the CAD, the implementation of actuators, the manufacturing and assembly of the real-life counterparts, and finally the choice and placement of the system's sensors, as well as the attachment of the finished design onto the human body. These design phases are described in further detail, chronologically in the following subsections of this chapter.

3.3.1 CAD design

The prototype was conceived in three mains pieces, these being:
One pipe like piece relating to the upper arm, seen in figure 4, where a rigid rotational link would be placed at the position of the elbow. This would be connected to the

servo motor housing, a structure provided during the course. Lastly, A mesh structure made of TPU was used to guarantee a more even force distribution as well as its compatibility with the FSR 402 set up.

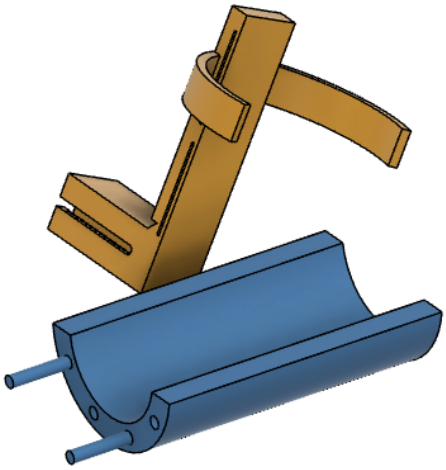


Figure 4: Upper arm piece in yellow and top part of forearm support in blue

The second piece refers to the forearm. It is depicted in figure 5. A continuation of the pipe like piece. These were connected with a slot mechanism with 4 supporting columns and 4 holes for attachment. This piece is responsible of supporting the forearm and leads to a ring. Such ring is connected to a 2:1 gear conversion that contains a 90mm ball bearing inside. This is the piece responsible for actuation translation into the final piece.

Lastly, a solid T-shaped piece was printed to act as the hand support as well as a basis through which a torque alongside the rotational axis for the wrist pronation and supination can be applied onto the hand. A Velcro strap is used to secure the limb onto the piece. The piece is bolted into the larger gear. As mentioned before, the gear ratio between the smaller gear and the larger gear is 2:1, as visible in figure 6 . Adding onto that, the RoM of both of the two servomotors is only 180°. This in result leads to a 90 degrees RoM for the wrist pronation and supination. This may be an issue if the goal were to design an exoskeleton that would make use of the human's whole RoM. In the observed use case, this is negligible, since picking up tasks here are happening usually within a RoM of 90°. This however does not pose a problem for the

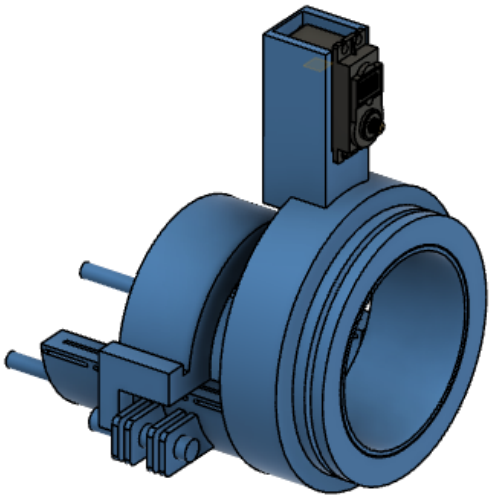


Figure 5: Bottom of the forearm piece and ball bearing in blue with servo housing and servo in black

elbow flexion and extension, whose DoF is less than 180°.

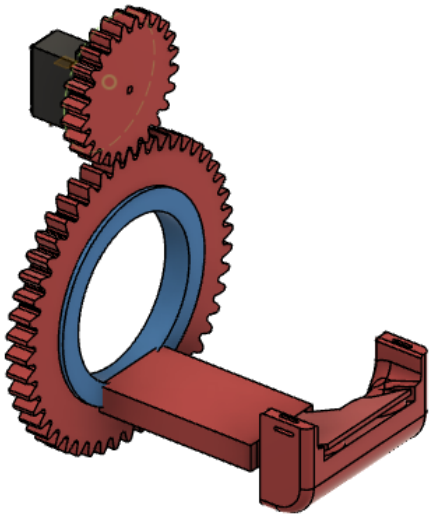


Figure 6: 2:1 Gear conversion in red, Ball bearing in blue and servo in black

3.3.2 Actuation

Considering the required torques for human upper limb assistance, two stronger servo motors compared to the ones in the tutorial are being used. Both motors have a maximum output torque of 40kg/m. The first acted on the elbow joint. This was linked directly,enabling it to encompass the full anatomical RoM of the elbow flexion and extension.

The second motor acted on the forearm, making use of the previously mentioned 2:1 gear conversion ratio. This arrangement was responsible for actuating the elbow pronation and supination and could have a a full, or limited RoM depending on the current assistive method used.

3.3.3 Manufacturing and Assembly

The pieces were manufactured via additive manufacturing, making use of extrusion printing of PLA via the devices "Prusa i3 MK3". This allowed for considerable freedom of design, which was necessary to fine tune the piece towards the use case. Limited amount of post machining was necessary due to miss-dimensioning. Such was accomplished by drilling, sanding and gluing the final product on assembly.

3.3.4 Sensor Placement

Considering that the main inputs of the system are standard fsr402 force sensors 2 pairs of spots for placing such were needed. With two sensors for a degree of freedom, it is possible to distinguish between the direction of movement alongside the rotational axis of the observed degree of freedom. The sensors hence need to be placed in a way, that only one of them is being pressed / actuated for one direction of rotational movement around a given DoF and vice versa the second one for the movement in the other direction.
The first pair was for the pronation / supination DoF, getting one force sensor in each extremity of the knuckles. This was done on the basis that when the patient wants to rotate his hand a torque would be induced on the sensor, such would then be translated

to the normal component of the force applied.
The second pair of sensors was positioned on the back of the upper arm where, when flexion was attempted, the triceps would push against the sensor. The complementary sensor position was determined experimentally to be better positioned on the bottom part of the forearm, where, when extension was attempted, a normal force would be transmitted to the FSR sensor.

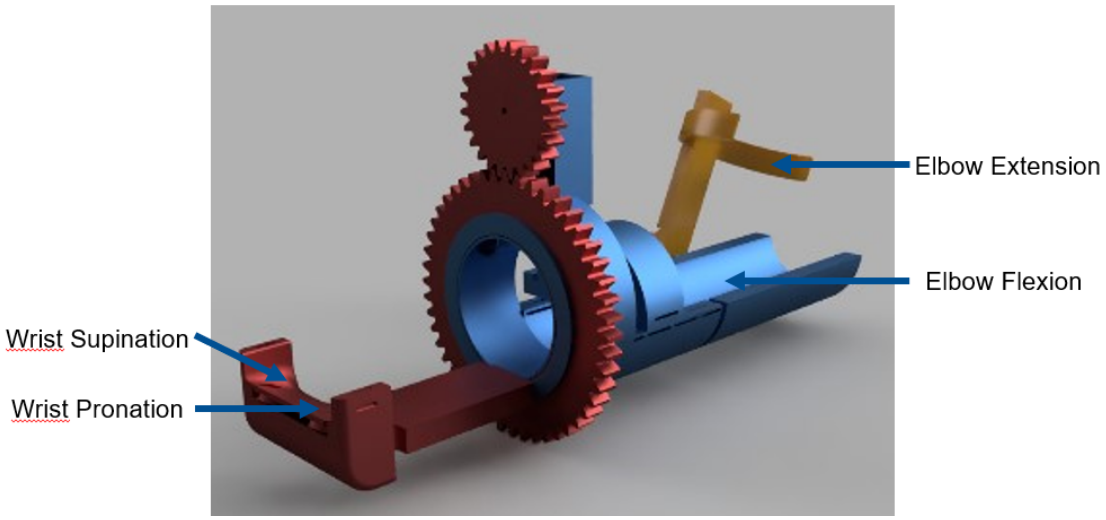


Figure 7: Placement of the used pressure sensors. For each degree of freedom 2 sensors are allocated to measure the human intent to perform a movement

3.3.5 Human Body Attachment

The exoskeleton is fixated on the human arm with the use of velcro straps as well as the aforementioned TPU mesh. The result is shown in figure 8 This set up allows for a flexible form-closure around the biceps and wrist. This not only allows the exoskeleton to be fixated firmly on the human's arm but also for the support of the human arm movement, both for pushing or pulling the arm alongside the rotation of each DoF. Furthermore, the better the exoskeleton is secured on the human arm, the more reliable the sensor inputs of the force sensors are, since their inputs are more reproducible.

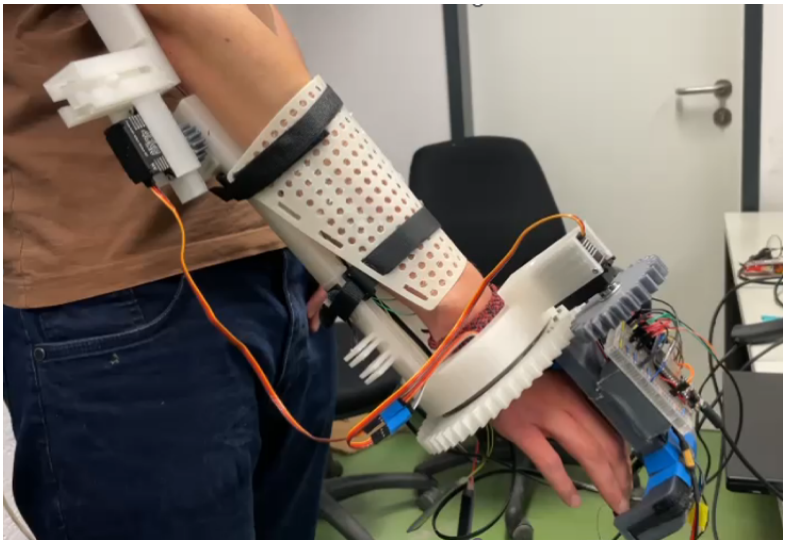


Figure 8: Resulting exoskeleton, attacked to the human arm using the TPU mesh and velcro straps

3.4 Process Model and Controller Design

The system is divided into two main parts: the Arduino Part, which is deployed on the ESP32, and the Python Part, which is deployed on the laptop, that the ESP32 is receiving its control inputs from. Both parts each handle specific tasks within the wearable exoskeleton's control system. While the Python Part is used to manage the controls and the assistive mode strategies, the Arduino Part is used to read the system inputs and apply the controller output onto the system. Both parts communicate with each other using a serial port connection. While the Python part sends the desired angles of the motors to the ESP32, the ESP sends the current sensor inputs to the Python part.

The ESP32 is responsible for control of both servo motors at wrist and elbow, each dedicated to the movement of their respective DoF. For both wrist and elbow control, the ESP32 module reads its analog and digital input pins, gathering inputs from the used force sensors and giving control inputs to the servomotors, according to the control input send from the python port. Before that however, the received angular values are being mapped according to a linear mapping function, in order to be readable by the servomotors. The mapping function is bounded by maximum and minimum values. This ensures, that movements of the motors are within the desired RoM, which in turn ensures the safety of the user by preventing the over extension of the human user's joints or any other dangerous maneuvers.

The Python part contains two separate parts, each of them being a program, which is running in parallel to its partner. Both programs are communicating with each other using pipelines. The first part is used to establish the communication between the ESP32 and the PC, on which the Python code is running. It is used to manage the sending and receiving of data packets from the PC to the ESP32 and vice versa. The data, received by the ESP32 is then sent to the second part of the Python part. This part manages the controls by transforming the acquired sensory data into a form that can be used for the controllers implemented here. There are two controllers implemented in the controller part of the Python code, one position controller and one force controller. Since the position controller is a feed forward controller, which outputs a trajectory, given a start and endpoint, it does not rely on the force sensor

readings. The force sensor readings are used for the force controller. Since the force controller computes its output using wrenches, the force sensor readings first need to be transformed from simple one-dimensional values to the desired wrench vectors which are each part of $\mathbb{R}^{6x1}$.

There are two assistive control methods implemented inside the exoskeleton controller part of the Python part, one of them being a continuous assistive mode and the other being an activity triggered assistive mode. The choice of mode is being done by toggling a physical switch on the breadboard, on which the ESP32 is located. Both control modes use the force controller. The activity triggered assistive mode on the other hand also uses the position controller. Depending on the use case, either the continuous assistance or the activity triggered assistance is more beneficial. The continuous assistance, by using the force controller aims to augment the user's force in response to the applied wrench on the joint, thereby strengthening the patient's physical capabilities. This functionality is crucial for both polymeric training and rehabilitation, as it supports the user by enhancing their force output progressively as movement continues. Conversely, in assistive Control mode, a position controller will be set on when the forces drop down below a threshold for a certain time, so that the joint will be brought to the desired given position. Since the position controller is set to always reach the maximum / minimum angles of the RoM (smallest or highest angular value), this mode is mostly beneficial for patients with movement impairment, who want to relearn specific movements and being able to access their full RoM again.

4 Evaluation and Future Work

In reviewing our current project, we see significant opportunities for improvement. Future efforts will focus on making our device better in several key areas.
It is aimed to reduce the amount of unnecessary movement during use, ensuring it's safer and more comfortable to operate as well as reducing the generated shear forces during movement. It is also suggested during trials, that the exoskeleton device itself should be stronger and more durable, so it can last longer and perform better in different conditions. Improving the placement of our sensors will make it to measure forces more accurately, which is crucial for the device to work as intended. Integration with EMG sensors to monitor muscle activity more closely, which could make the device even more responsive and effective. Additionally, the dynamic model could still be fine-tuned to make movements feel more natural to use. Finally, a combination of PID control and force control could help the device's movements more precisely.

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